

NL2TL-Tree: Natural Language to Nested Temporal-Logic Trees for Robustness-Guided Flow-Matching Planning

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Abstract

Letting non-expert users command robots in plain language is a long-standing goal of ubiquitous computing, but free-form instructions are ambiguous and give no guarantee that the resulting behavior is correct. We present **NL2TL-Tree**, an *early-stage prototype* exploring whether a *nested Signal Temporal Logic (STL) tree* can serve as an interpretable, checkable interface between natural-language robot instructions and motion planning. The pipeline is *training-free* (no model is trained or fine-tuned): a frozen LLM is few-shot-prompted to parse a smart-device task into the tree (human-readable and error-localizable to a single sub-node), and a *frozen* graph-encoded flow-matching planner (TeLoGraF), steered by a differentiable STL-robustness term, turns the tree into motion. Test-time mechanisms (operator normalization, nested-tree decomposition, a feasibility self-check that defers to a classical backend) extend the frozen planner, and an online robustness monitor can trigger a stop/re-plan; a *planning-latency-aware* deadline further debits the planner’s own wall-clock from timed goals. We report *preliminary evidence* (a small parsing pilot and a TurtleBot3 Gazebo simulation) and outline an evaluation plan for parsing accuracy, failure localization, and user-facing interpretability. Code, prompts, and configs are released.¹

CCS Concepts

• **Computing methodologies** → **Robotic planning**; Natural language generation; • **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

natural language interfaces, signal temporal logic, flow matching, diffusion models, training-free LLM parsing, task planning, verifiable AI, human-robot interaction

¹<https://github.com/gjc823878198/NL2TL-Tree-Natural-Language-to-Nested-Temporal-Logic-Trees-for-Robot-Flow-Matching-Planning>

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1 Introduction

Pervasive and wearable systems increasingly let people issue tasks in everyday language: “check the kitchen every ten minutes, but never enter the bedroom.” Such instructions are inherently *temporal* and *safety-constrained*, yet most natural-language interfaces translate them with end-to-end models whose output cannot be inspected, checked, or trusted before execution. For ubiquitous computing, where the system acts in a user’s physical space, this lack of interpretability and verifiability is a core barrier to adoption.

Signal Temporal Logic (STL) is an attractive target: it expresses temporal and spatial constraints precisely and admits a quantitative *robustness* of satisfaction. Two threads have grown around it (translating language into formulas [2, 3], and generative planning for STL tasks [7, 9]), but usually in isolation: language work stops at a formula, planning work assumes one is given.

We connect them with one modular pipeline (Fig. 1) built around a single idea: *keep an explicit, verifiable logical structure between the human and the planner*. The entry point is a smart device (e.g., a phone): a user types (or dictates) a task and receives a checked *spatiotemporal command* (where to be, what to avoid, when) that a mobile robot executes. The STL spec enters a frozen flow-matching planner natively through its graph encoder, and a differentiable robustness term steers sampling; nothing is trained, so the train/use gap is closed at test time (§4).

In this poster we present the NL2TL-Tree *prototype* (NL → nested STL tree → robustness monitor → frozen flow-matching planner) built around the nested tree as an *interpretable, checkable* intermediate representation, and report a *preliminary feasibility study* (a frozen-LLM parsing pilot and a Gazebo simulation, §4). The structural-accuracy levers (self-consistency, round-trip and robustness-feedback self-correction) and a user study are *ongoing*, not yet evaluated.



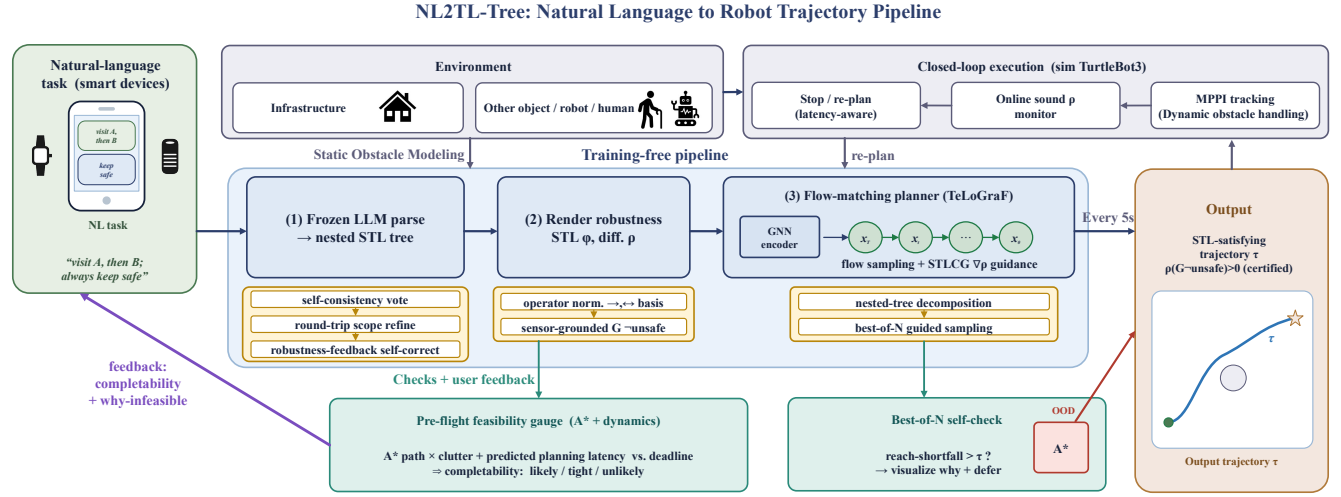


Figure 1: The NL2TL-Tree pipeline: everything frozen / training-free. A smart-device task is parsed by a frozen LLM into a nested STL tree, rendered to a differentiable robustness ρ , and planned by a frozen flow-matching planner (TeLoGraF) under STL_{CG} gradient guidance. The trajectory executes closed-loop on a simulated TurtleBot3: the planner models sensed obstacles *statically*, a reactive MPPI tracker avoids the *live*-sensed field, and an online sound- ρ monitor triggers latency-aware stop/re-plans. Each stage is annotated with its test-time mechanisms (§4); a *checks-and-feedback* layer returns the completability and a *why-infeasible* explanation to the user.

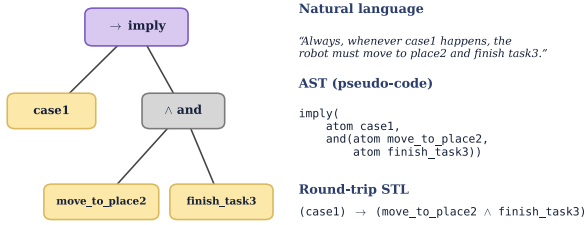


Figure 2: The interpretable intermediate representation. A natural-language instruction (top right) is parsed into a nested STL tree (left); its JSON abstract syntax (right) is what downstream stages consume, and a round-trip back to an STL string (bottom right) aids verification. The tree exposes operator scope: the structure the parser must get right and that self-correction repairs.

2 Approach

2.1 Training-free natural language to a nested STL tree

Rather than emitting a flat formula string, we prompt a *frozen* LLM $\mathcal{M}$ to produce a *nested logical tree* serialized as JSON (Fig. 2): atomic predicates at the leaves, Boolean/temporal operators ($\square$, $\diamond$, $\mathcal{U}$, timed variants) at internal nodes. Unlike a flat formula, this representation is *human-readable*, *grammar-checkable* before execution, and *error-localizable* to a single sub-tree, and a round-trip back to an STL string aids verification (Fig. 2, bottom).

By *training-free* we mean *parameter-update-free*, not data-free: no model weights change; the LLM is steered only by K_{shot} in-context

exemplars and the planner stays frozen. The LLM is invoked with a strict-JSON output mode and a prompt assembled from K_{shot} input-output exemplars; we treat K_{shot} as a user-tunable budget ($K_{\text{shot}}=8$ here; larger values trade tokens for coverage of operator diversity). Schema validation (VALIDATESchema) rejects malformed JSON or unknown operators before downstream use.

Robustness-feedback self-correction. Rather than accept the first parse, we *verify* it and re-prompt the same frozen LLM when a check fails: a neuro-symbolic loop with no parameter updates. Three verifiers fire cheapest-first: (1) operator arity/schema; (2) grounding consistency (every atom names a known place, no place is both reach and avoid); and (3) *quantitative STL robustness*: we plan a trajectory and, if its robustness ρ cannot be made positive, the spec is unsatisfiable/contradictory and we feed ρ and the offending sub-tree back as a targeted correction. Each failed round appends the rejected JSON plus the diagnostic to the prompt and re-queries (bounded rounds). This directly attacks our dominant error mode (right operators, wrong nesting; see §4); it is implemented, but its quantitative gain is pending the larger study.

Structural-accuracy levers (training-free). Since the residual error is almost purely *structural* (§4), we add two mechanisms that target the nesting. (i) **Structural self-consistency**: sample K parses at temperature > 0 , canonicalize each tree, and majority-vote the structure [13]: operators barely vary across samples while scope does, so the majority recovers nesting that greedy decoding loses to a single near-miss. (ii) **Round-trip scope refinement**: the tree is paraphrased back to NL with its scope made explicit and regenerated only when it disagrees with the instruction [10], turning an invisible inversion into a checkable one.

Role of the public NL-STL dataset. We use a public NL-to-STL dataset three frozen ways: a *few-shot pool* (K_{shot} exemplars); a source of *gold trees* (formulas converted once off-line by a Lark grammar into evaluation-only references); and a *reservoir* of mis-parses written back as future exemplars: none trains a model.

2.2 Rendering to differentiable robustness

The tree compiles to a canonical STL formula φ and a *differentiable robustness* $\rho(\cdot; \varphi)$ under the standard quantitative semantics ($\rho(\neg) = -\rho$, $\rho(\wedge) = \min$, $\rho(\diamond) = \max$, dually for $\Box, \mathcal{U}$; $\rho > 0 \Leftrightarrow x \models \varphi$; Fig. 3). Following masking-based formulations [8], this smooth ρ is an *optimizable surrogate* that steers sampling; a *separate sound monitor* (exact, non-smoothed) does the validation, with a *duration-aware* keep-safe robustness [6] (worst-case clearance on a satisfied path; penalizing unsafe duration under violation). Optimizer and judge are different functions. We anchor the index axis to wall-clock seconds (arclength at a nominal speed), so $F_{[0,D]} \text{REACH}$ is a real *deadline*; the monitor validates keep-safe and reach-by-deadline on the executed trace. Both margins are in *metres*: ρ_{safe} is worst-case clearance, ρ_{time} the robustness of $F_{[0,D]} \text{REACH}$ within the deadline window (>0 iff the goal is entered before D). Run *online*, a non-positive ρ stops and re-plans (a runtime monitor in simulation), so a violation is repaired, not merely logged.

Planning-latency-aware deadline. The flow-matching planner is not instantaneous (cold-start, then seconds per re-plan), and that wall-clock elapses against $F_{[0,D]}$: the naive ρ_{time} *freezes* time while the planner thinks. We predict its latency online (cold-start-seeded EMA) and *debit* it: $\rho_{\text{time}}^{\text{lat}} = (D - \hat{t}_{\text{plan}}) - t_{\text{exec}} \leq \rho_{\text{time}}$, handing the tightened deadline to the next re-plan; a plan that “arrives in time” ignoring its own planning cost is then flagged late.

Sensor-grounded keep-safe. Obstacle avoidance is *one* clause $\Box \neg \text{UNSAFE}$, with UNSAFE grounded by the union of LiDAR-sensed disks $\{(c_k, r_k)\}$, so $\rho(\Box \neg \text{UNSAFE}) = \min_t \min_k (\|p_t - c_k\| - r_k)$ is the worst clearance. A union predicate (vs. N avoid atoms) keeps the GNN encoder in-distribution while the *same* sensed set grounds the keep-safe semantics and drives guidance, so what the robot senses becomes its keep-safe constraint, monitor-validated online ($\rho > 0$) under the sensed obstacle model.

2.3 Tree to trajectory: a frozen flow-matching planner

We pair ρ with a *frozen* generative planner: concretely TeLoGraF [12], which ingests the STL graph through a GNN encoder (the spec enters natively as conditioning) and samples trajectories by flow matching; a differentiable ρ term steers the sampler toward high-satisfaction regions (Fig. 3). The recipe also fits score-based backbones [1, 4, 7, 11]; weights are never updated. **Three test-time mechanisms** make the frozen planner usable on real specs (Fig. 3), all leaving p_θ untouched. (i) **Operator normalization**: the graph vocabulary lacks $\rightarrow, \leftrightarrow$, so we rewrite them with the standard identities $A \rightarrow B \equiv \neg A \vee B$ and $A \leftrightarrow B \equiv (A \rightarrow B) \wedge (B \rightarrow A)$, verified to preserve robustness *exactly* (zero deviation over thousands of traces and a Boolean truth-table check). (ii) **Nested-tree decomposition**: a nested sequential spec is out-of-distribution, yet each leg is an in-distribution timed reach, so we split the tree, flow-match each leg, and stitch (cf. [5, 9]). (iii) **Feasibility self-check**: from the spread

of the best-of- N prior’s reach-shortfall (an *empirical* OOD signal that separates our flow-feasible specs from the out-of-distribution nested one), we defer to a classical planner only when even the best candidate is beyond guidance’s reach (best-of- N per-atom reach-shortfall $> \tau_{\text{OOD}} = 2.5$ m), making the generative-vs-classical choice *measured*, not fixed. The same A^* /dynamics estimate also gives a **pre-flight completability** check (an A^* path through the goals, inflated for clutter and *charged the predicted planning latency*, timed against the deadline), flagging an unreachable deadline up front rather than mid-run.

3 Design Rationale

The pipeline is *designed* for four structural properties, though whether they yield measurable *user* benefits (faster error-spotting, calibrated trust) is a hypothesis we have *not yet user-validated* (§5). It is *interpretable* (the human-planner interface is an explicit nested tree, so a wrong instruction localizes to one sub-node), *checkable* (the sound monitor of §2.2 checks the executed run at the monitor level, $\rho \leq 0$ triggering a stop/re-plan in simulation, with intent aligned by the round-trip check), *training-free* (LLM and planner stay frozen), and *modular* (the typed $\text{NL} \rightarrow \text{tree} \rightarrow \rho \rightarrow \text{trajectory}$ interfaces let any stage be swapped). Unlike NL2TL [3], which fine-tunes a model and stops at a *flat* formula, NL2TL-Tree is training-free, emits a *checkable nested tree*, and carries the spec through to execution; and unlike the planner we reuse (TeLoGraF [12], which assumes the STL is given and is tied to a fixed operator set), we keep it *frozen* and extend it purely at test time.

The user-facing loop. The tree is the interaction surface (Fig. 2): a user types a task on a smart device, sees the parsed nested tree rather than an opaque trajectory, localizes a mis-scoped constraint to one sub-node, edits or re-prompts, and reads a keep-safe / reach-by-deadline *monitor* value ($\rho > 0$, a monitor-level check) *before* the robot moves: an *interactive, checkable command interface* rather than a black box. Whether it speeds error-spotting for non-experts is left to a formative study (§5).

4 Case Study and Evaluation

We report a *preliminary* case study (not a full benchmark): a parsing study with a frozen LLM (Stage 1) and an end-to-end planning evaluation in TurtleBot3 *simulation* (Gazebo Classic) (Stage 2).

4.1 Stage 1: NL $\rightarrow$ STL tree (frozen LLM)

Data + protocol. We use the public NL2TL “lifted” dataset (predicates abstracted to $\text{prop}_1, \dots$), covering all eight STL operators and their unbounded variants [3]; each row pairs a sentence with a gold STL formula, which a Lark grammar converts once into a gold tree. We call the frozen LLM in strict-JSON mode (zero training; full settings in the released configs); it never sees the formula, and its predicted tree is scored directly against the gold tree. We score the first 30 distinct sentences (no cherry-picking; every parse, including errors, is counted).

Metrics. Beyond strict equality: *op-F1* (operator multiset: right pieces?), *path-F1* (root-to-leaf path multiset: right nesting?), and normalized tree-edit distance ($1 - \text{TED} / \max(|T_p|, |T_g|)$).

Result (preliminary pilot, $n=30$, no fine-tuning). As an initial feasibility check, the frozen LLM *usually picks the right operators*

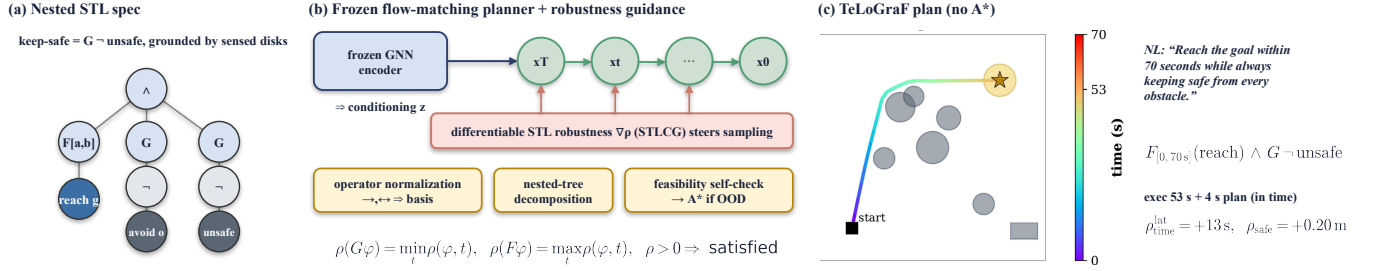


Figure 3: From spec to trajectory (§2.2–2.3). (a) the nested STL tree, with reach (blue) and avoid / keep-safe (grey) atoms; (b) a frozen GNN encoder conditions a flow-matching sampler ($x_T \rightarrow x_0$) steered by $\nabla \rho$, with the three test-time mechanisms; (c) a timed task solved by *pure* frozen TeLoGraF (no A*): the trajectory, coloured by wall-clock seconds, enters the goal at 53 s of a 70 s deadline (latency-aware $\rho_{\text{time}}^{\text{lat}} = +13$ s, keep-safe $\rho_{\text{safe}} = +0.20$ m).

Table 1: Per-task Stage-2 outcome on the cluttered scene of Fig. 3 (six hand-built tasks). On all six the sound monitor reports keep-safe $\rho(\Box \neg \text{UNSAFE}) > 0$ and goal entry; ρ_{safe} is the mean keep-safe margin over 3 re-runs. The engine is selected automatically (“flow” = pure TeLoGraF + STLCG, “+decomp.” = per-leg flow, “A*” = classical fallback); the lone out-of-distribution spec is routed to A* by design, not a silent success.

Task	STL template	Planner	ρ_{safe} (m)
reach_within_T	$F_{[a,b]} g \wedge \Box \neg \cup$	flow	+0.20
reach_avoid	$F g \wedge \Box \neg o$	flow	+0.20
reach_goal_north	$F g \wedge \Box \neg o$	flow	+0.20
seq_reach_ABC	$F(A \wedge F(B \wedge F C))$	decomp + A*	+0.27
cond_reach_either	$\neg F N \rightarrow F S (\equiv F N \vee F S)$	flow	+0.20
trigger_response	$\Box(p \rightarrow F_{[0.5]} q)$	A* (OOD)	+0.21

(op-F1=0.95) but *mis-nests scope* (path-F1=0.64); exact-match is 56.7% (95% Wilson CI [39, 73] %: wide, as expected for a pilot) and the average tree is only 1.03 edits from gold ($\text{TED}_{\text{norm}}=0.84$). The error is thus almost entirely *structural*, which motivates the structural self-consistency voting [13] and round-trip scope refinement [10] levers. A small ablation ($n=30$): self-consistency ($K=5$) helps modestly (exact 0.57→0.60), while round-trip refinement did *not* (it lowered exact-match). These pilot figures are indicative, pending a larger evaluation.

Flat-STL baseline. A flat-STL baseline (the same frozen LLM emits an STL *string*, parsed by the same grammar; same sentences/gold/metrics) is *no more accurate*: across two further open LLMs the tree matches or beats it on op-F1 (0.94–0.96 vs. 0.90–0.92) and path-F1, trailing only slightly on exact-match (0.60–0.63 vs. 0.67, both grammar-canonicalized). The interpretable structure is essentially free.

What the study stresses. These pilot scores matter less than the *test-time machinery* of §2; the *why-infeasible* feedback and the best-of- N self-check visualization run live in our demo supplement.

4.2 Stage 2: tree → trajectory

We evaluate the frozen-TeLoGraF planner two ways, as an illustrative case study rather than a generalization benchmark.

(i) *Per-task study (cluttered scene; Table 1, Fig. 3(c)).* On six hand-built tasks spanning the STL templates, the sound monitor reports $\rho(\Box \neg \text{UNSAFE}) > 0$ and goal entry. An *ablation* (3 re-runs/task): pure flow + STLCG solves the four unnested timed-reach tasks (4/6) but not the nested sequential or OOD response; the full pipeline recovers both (6/6), and disabling the A* repair drops back to 4/6: the *repair is the load-bearing recovery*, decomposition alone leaving the count unchanged. Operator normalization separately lets pure flow discharge an implication spec (cond_reach_either). (ii) *Simulated closed loop (TurtleBot3, Gazebo Classic; full setup in the released configs).* On live LiDAR it reaches the goal in 35/53 s ($\rho_{\text{time}} = +0.1$, keep-safe +1.2 m); a *multi-goal* variant visits three regions in order through a cylinder field, all in time.

Metrics. Beyond satisfaction and robustness ρ , we report *inference time*: each re-plan completes in 3.6–4.3 s (mean 3.9, $n=6$) on a laptop CPU, the very latency our planning-aware $\rho_{\text{time}}^{\text{lat}}$ debits (§2.2). Failures are detected, not hidden: the post-hoc check routes a failing plan to A* repair.

Planned baselines. A study (*not yet run*) will compare against classical LLM → STL → solver [2], generative [1, 7], and STL/LTL-aware [4, 5, 9, 11, 12] planners.

5 Discussion and Future Work

The structured, training-free middle layer is a promising abstraction for NL robot control: the nested tree is *checkable* and error-localizable to one sub-node, at no parsing-accuracy cost in our pilot. The system composes existing frozen components (LLM, STLCG, TeLoGraF, A*): the contribution is the abstraction and its test-time mechanisms, not any single component (§4 attributes the gains). The six tasks establish feasibility, not superiority; interpretability and trust benefits remain hypotheses until user-validated. **Future work** spans a larger parsing benchmark; the baseline study of §4 with a factorial ablation; a user study of whether non-experts detect and repair mis-parses faster from the tree than from a flat formula; real-LiDAR hardware; and a better-matched TL robustness.

Acknowledgments

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